- A change in the methodology used to calculate the debt indicators from the HIPC DSA framework to the Low Income Countries DSA framework (DSF). These changes are reflected in the definition of exports, the discount rate used to calculate the NPV of debt and the exchange rates used to calculate the debt service projections.[20]
- HIPC completion point is currently assumed for mid-2006. By contrast, the decision point document assumed the attainment of HIPC completion point in 2003.
- Discount rates used to estimate NPV values in the current analysis are lower than those used in 2000.
- São Tomé and Príncipe's export performance since 2000 has been much weaker than expected.
- There has been some unanticipated new external borrowing, due to a more expansionary fiscal stance than originally envisaged.
- An unanticipated depreciation of the US dollar against the euro since 2000.

VIII. DEBT SUSTAINABILITY ANALYSIS

7. **The macroeconomic framework in the staff baseline scenario (Box 3)[21] assumes a steady increase in economic activity over the medium term, supported by sound macroeconomic policies and the development of the domestic oil sector.** Domestic investment would rise sharply from about 36 percent of GDP to an average of 54.7 percent of GDP in 2008–10, largely on account of oil sector investment initiatives. The rise in the investment ratio will require a continuous reliance on foreign savings, although national savings are forecast to rise on account of an improved financial position of the government and increased private savings once stability and economic growth become firmly established. Key risks to the macro framework arise from a possible weakening of the fiscal consolidation effort in the wake of elections scheduled for 2006. Also, potential shortfalls in foreign assistance triggered by the future oil boom could exacerbate the government's budget constraint in the short run.

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[20] The DSF uses same-year exports, as opposed to the 3-year backward average used for the decision point. Regarding the discount rates, the DSF methodology uses a fixed 5 percent rate compared to the currency-specific discount rates under the HIPC methodology. Finally, debt service is projected using WEO assumptions in the DSF, as opposed to fixed (end-1999) exchanges rates used for the decision point. See: "Debt Sustainability in Low-Income Countries: Proposal for an Operational Framework and Policy Implications"www.imf.org and IDA/SecM2004–0035 and "Debt Sustainability in Low-Income Countries—Further Consideration on an Operational Framework and Policy Implications,"www.imf.org and IDA/SecM2004–0629/1.

[21] The baseline scenario assumes the attainment of HIPC completion point by mid-2006.

Box 3: São Tomé and Príncipe: Macroeconomic Assumptions in Baseline Scenario

Real GDP growth is projected to average 6 percent in 2005–10 reflecting the implementation of the PRSP and continuous donors' support. The average growth rate is projected to increase to 8 percent in 2011–25 on account of increasing oil production.

Exports of goods and services are projected to increase at an average rate of 10 percent during 2005–10 as a result of strong growth in tourism and a gradual recovery of cocoa export volumes. Oil exports explain the strong export performance projected for 2011–25.

The external current account deficit is expected to average 30 percent of GDP in 2005–10; i.e., roughly unchanged with respect to recent years. The current account is projected to turn into a surplus of about 30 percent of GDP in 2011–25 due mainly to oil exports.

Fiscal projections assumes overall fiscal deficits declining from 18 percent of GDP on average in 2005–06 (excluding oil signature bonuses) to 6 percent in 2010, and a balanced budget thereafter, largely reflecting oil tax receipts.

Multilateral external loan financing is projected at an average of about US$7.5 million per year during 2005–09, leveling off to US$6 million in 2010–25.

8. **In the baseline scenario, São Tomé and Príncipe's debt outlook shows gradual improvements over the medium term, but it will remain a source of vulnerability until oil production starts in 2012.** The NPV of public debt-to-GDP ratio falls from 153 percent in 2005 to 110 in 2006, and to 92 percent in 2010, reflecting a prudent public expenditure envelope underpinning the government's medium-term fiscal strategy (Table 1a).[22] Only in 2012, the ratio falls significantly to 22 percent of GDP, due mainly to a large increase in GDP associated with the start in oil production. The NPV of debt-to-revenue ratio, at 278 percent in 2005 (Figure 1 and Table 1b), improves significantly in 2006 and remains around 160 percent during 2007–12, before declining gradually thereafter. The debt service-to-revenue ratio drops significantly following HIPC completion point in 2006 and remains sustainable thereafter.

9. **The debt dynamics are particularly vulnerable to shocks**. Sensitivity analysis indicates that under the most extreme stress test, defined as a 30 percent depreciation of the dobra exchange rate (measured in U.S. dollar terms), the NPV of debt-to-GDP ratio increases to 186 percent in 2006, compared to about 110 percent in the baseline scenario (see Figure 1 and Table 1b). This reflects the fact that São Tomé and Príncipe's public debt is denominated

[22] All public debt is owed to foreign residents (i.e., external public debt).

in foreign currency. The scenario of no-fiscal-reform (defined to include an unchanged primary fiscal balance from its 2004 level, in terms of GDP) brings about an explosive debt burden, underlying the importance of fiscal consolidation to improve the external debt outlook.

IX. External Debt Sustainability

10. **The NPV of debt-to-exports ratio improves steadily in the baseline scenario, but will remain significantly above the policy-dependent thresholds until oil production starts.** The NPV of debt-to-export ratio decreases from 500 percent in 2005 to 315 percent in 2006, to 271 percent in 2010 (Table 2b). The ratio falls significantly below the 100 percent policy-dependent debt-burden threshold only in 2012, when oil production and exports are projected to begin. The debt service-to-exports ratio declines from 56 percent[23] in 2005 to about 11 percent during 2007–10, before declining steadily during the following decade, reflecting the delivery of HIPC debt relief and the high concessionality in which new debt is contracted.

11. **Sensitivity analysis shows that the NPV of debt-to-exports ratio is particularly vulnerable to a deceleration in export growth.** Specifically, a decline in the export growth rate by one standard deviation from historical levels increases the ratio assessed at 300 percent under the baseline scenario to 424 percent in 2007 (Figure 2 and Table 2b).

X. The Effects of MDRI

12. **The implementation of the MDRI would help reduce São Tomé and Príncipe's debt vulnerability (Figure 3, Table 3, and summary table below).** The MDRI would reduce the NPV of debt-to-exports ratio from 315 percent in the original baseline scenario to 176 percent in 2006, under the modified baseline; roughly 70 percent higher than the sustainable level in the case of a poor performing country. While NPV ratios would remain above the policy dependent thresholds, the debt service-to-exports ratio would be cut significantly during 2006–10 under the MDRI resulting in substantial savings. The latter reflects the relatively large share of debt to the World Bank and the African Development Bank in the country's total debt.

[23] In 2005, debt service refers to debt service due before HIPC completion point and excludes HIPC interim debt relief, rescheduling and cash payments on account of arrears.

São Tomé and Príncipe: DSA Baseline and MDRI Scenarios 1/
(In percent)

	Est.	Projections						
	2005	2006	2007	2008	2009	2012	2015	2025
				NPV of debt-to-GDP ratio				
Before MDRI	153	110	106	102	97	22	7	10
After MDRI	. . .	61	62	61	60	14	5	7
				NPV of debt-to-exports ratio				
Before MDRI	500	315	302	289	282	46	7	20
After MDRI	. . .	176	176	174	175	30	5	14
				Debt service-to-export ratio 2/				
Before MDRI	56.0	19.1	12.4	11.4	10.6	2.4	0.4	2.1
After MDRI	. . .	9.9	6.7	6.5	6.0	1.8	0.3	1.3
				Debt service-to-GDP ratio				
Before MDRI	17.1	6.7	4.4	4.0	3.6	1.2	0.4	1.1
After MDRI	. . .	3.5	2.4	2.3	2.1	0.8	0.3	0.7

Source: Staff projections and simulations.
1/ Baseline assumes Paris Club rescheduling in 2005 and HIPC completion point by mid-2006.
2/ Refers to exports of goods and nonfactor services.

XI. CONCLUSION

13. **As noted in the latest World Bank Country Assistance Strategy (IDA/R2005-0087, May 2005) and IMF staff report (IMF Country Report No.05/323, July 18, 2005), the country faces a high risk of debt distress until oil production starts early in the next decade.** Also, São Tomé and Príncipe's public debt will remain high and vulnerable to shocks to the exchange rate, exports and foreign grants, at least until oil production starts in 2012/13. Debt vulnerability would be further exacerbated in the absence of sound macroeconomic policies and undue delays with oil production. Calculations indicate that the country would benefit from a significant reduction in debt service under the MDRI starting in 2006.

Table 1a.São Tomé and Príncipe: Public Sector Debt Sustainability Framework, Baseline Scenario, 2002–2025
(In percent of GDP, unless otherwise indicated)

	Actual			Estimate	Projections										
	2004	Historical Average 5/	Standard Deviation 5/	2005	2006	2007	2008	2009	2010	2005-10 Average	2011	2012	2015	2025	2011–25 Average
Public sector debt 1/	467.9			419.7	355.4	328.1	301.3	275.2	249.2		224.2	54.9	16.3	19.9	
Of which : foreign-currency denominated	467.9			419.7	355.4	328.1	301.3	275.2	249.2		224.2	54.9	16.3	19.9	
Change in public sector debt	-23.8			-48.3	-64.3	-27.3	-26.8	-26.1	-26.0		-25.1	-169.3	-0.7	0.0	
Identified debt-creating flows	-10.8			10.6	-10.0	-23.5	-24.5	-23.8	-22.5		-28.8	-169.1	-1.0	-0.1	
Primary deficit	21.8	11.8	8.1	13.5	14.6	9.7	5.7	4.6	4.3	8.7	-2.0	-0.5	-0.1	-0.2	-0.3
Revenue and grants	60.6			54.8	61.1	65.3	64.0	60.8	57.3		50.6	13.2	8.6	24.6	
Of which : grants	32.2			24.6	29.0	32.8	31.9	28.9	25.5		21.0	5.2	0.7	0.3	
Primary (noninterest) expenditure	82.4			68.3	75.7	75.0	69.7	65.3	61.6		48.6	12.7	8.5	24.4	
Automatic debt dynamics	-25.2			3.9	-24.5	-33.2	-30.1	-28.4	-26.8		-26.8	-168.7	-0.9	0.1	
Contribution from interest rate/growth differential	-25.0			-23.2	-21.7	-22.4	-23.8	-23.2	-22.9		-21.6	-19.1	-1.5	-1.7	
Of which : contribution from average real interest rate	-7.0			-6.1	-3.6	-3.9	-3.7	-3.5	-3.7		-3.1	-2.5	-0.2	-0.2	
Of which : contribution from real GDP growth	-18.0			-17.1	-18.1	-18.5	-20.0	-19.7	-19.2		-18.5	-16.6	-1.3	-1.5	
Contribution from real exchange rate depreciation	-0.2			27.2	-2.8	-10.8	-6.3	-5.2	-3.9		-5.2	-149.5	...	...	
Other identified debt-creating flows	-7.4			-6.8	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	
Privatization receipts (negative)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	
Recognition of implicit or contingent liabilities	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	
Debt relief (HIPC and other)	-7.4			-6.8	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	
Residual, including asset changes	-13.0			-58.9	-54.4	-3.8	-2.4	-2.3	-3.5		3.8	-0.1	0.3	0.0	
NPV of public sector debt	156.3			152.6	110.1	106.4	101.6	96.7	92.2		85.6	21.6	6.9	10.2	
Of which: foreign-currency denominated	156.3			152.6	110.1	106.4	101.6	96.7	92.2		85.6	21.6	6.9	10.2	
Of which: external	156.3			152.6	110.1	106.4	101.6	96.7	92.2		85.6	21.6	6.9	10.2	
NPV of contingent liabilities (not included in public sector debt)	...			...	...	...	...	...	...		...	...	...	...	
Gross financing need 2/	38.2			30.6	21.2	14.1	9.7	8.2	7.4		2.1	0.7	0.3	0.9	
NPV of public sector debt-to-revenue ratio (in percent) 3/	258.0			278.5	180.1	163.0	158.7	159.2	161.0		169.0	163.6	80.2	41.4	
Of which: external	258.0			278.5	180.1	163.0	158.7	159.2	161.0		169.0	163.6	80.2	41.4	
Debt service-to-revenue ratio (in percent) 3/ 4/	27.1			31.2	10.9	6.7	6.2	6.0	5.4		8.1	8.7	4.8	4.3	
Primary deficit that stabilizes the debt-to-GDP ratio	45.6			61.8	78.9	37.0	32.5	30.6	30.3		23.1	168.8	0.6	-0.1	
Key macroeconomic and fiscal assumptions															
Nominal GDP (local currency)	629			749	889	1050	1219	1406	1608		1813	7363	24273	16640	
Real GDP growth (in percent)	3.8	2.8	1.1	3.8	4.5	5.5	6.5	7.0	7.5	5.8	8.0	8.0	8.0	8.0	8.0
Average nominal interest rate on forex debt (in percent)	1.1	1.9	1.4	1.1	1.1	0.8	0.8	0.7	0.5	0.9	0.9	0.9	0.9	0.9	0.9
Average real interest rate on domestic currency debt (in percent)	...	3.3	7.7	...	...	...	...	...	...		...	...	...	...	
Real exchange rate depreciation (in percent, + indicates depreciation)	0.0	15.4	46.4	6.1	...	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	9.7	5.2	4.1	14.7	13.6	11.9	9.1	7.8	6.4	10.6	4.4	276.1	-4.2	-9.8	21.4
Growth of real primary spending (deflated by GDP deflator, in percent)	21.5	43.9	58.1	-13.9	15.8	4.5	-1.0	0.3	1.4	1.2	-14.7	-71.8	56.0	18.4	8.4

Sources: Country authorities; and Fund staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

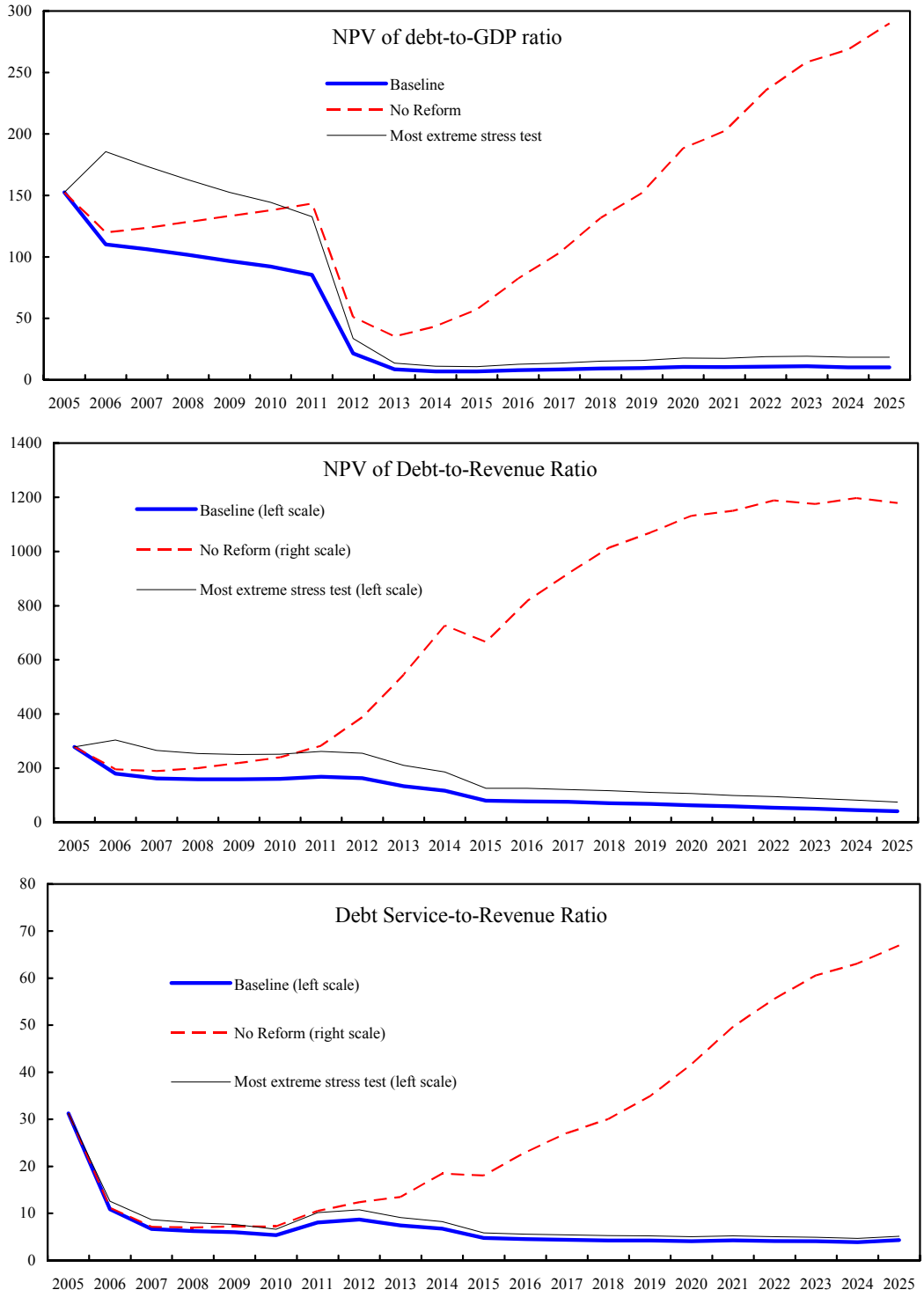
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 1b.São Tomé and Príncipe: Sensitivity Analysis for Key Indicators of Public Debt 2005–2025

	Est.	Projections								
	2005	2006	2007	2008	2009	2010	2011	2012	2015	2025
NPV of Debt-to-GDP Ratio	**NPV of Debt-to-GDP Ratio**									
Baseline	153	110	106	102	97	92	86	22	7	10
A. Alternative scenarios										
A1. Real GDP growth and primary balance are at historical averages	153	110	110	113	116	120	126	42	39	246
A2. Primary balance is unchanged from 2004	153	120	124	128	133	138	143	51	57	290
A3. Permanently lower GDP growth 1/	153	110	107	103	98	94	88	22	7	16
B. Bound tests										
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	153	114	115	111	107	104	97	25	9	22
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	153	114	116	111	106	101	93	24	8	11
B3. Combination of B1-B2 using one half standard deviation shocks	153	113	116	110	103	97	90	22	7	10
B4. One-time 30 percent real depreciation in 2006	153	186	174	163	153	144	133	34	11	19
B5. 10 percent of GDP increase in other debt-creating flows in 2006	153	117	112	107	102	97	90	23	7	11
NPV of Debt-to-Revenue Ratio 2/	**NPV of Debt-to-Revenue Ratio 2/**									
Baseline	278	180	163	159	159	161	169	164	80	41
A. Alternative scenarios										
A1. Real GDP growth and primary balance are at historical averages	278	178	165	169	180	194	227	284	437	984
A2. Primary balance is unchanged from 2004	278	196	189	201	219	241	283	388	666	1179
A3. Permanently lower GDP growth 1/	278	180	164	160	161	163	172	168	86	66
B. Bound tests										
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	278	184	171	169	171	176	187	184	101	89
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	278	186	178	173	174	175	184	178	87	46
B3. Combination of B1-B2 using one half standard deviation shocks	278	183	174	167	166	166	173	167	82	42
B4. One-time 30 percent real depreciation in 2006	278	304	266	254	251	252	262	255	126	75
B5. 10 percent of GDP increase in other debt-creating flows in 2006	278	191	172	168	168	170	178	173	85	44
Debt Service-to-Revenue Ratio 2/	**Debt Service-to-Revenue Ratio 2/**									
Baseline	31	11	7	6	6	5	8	9	5	4
A. Alternative scenarios										
A1. Real GDP growth and primary balance are at historical averages	31	11	7	6	6	6	10	11	14	62
A2. Primary balance is unchanged from 2004	31	11	7	7	7	7	11	12	18	67
A3. Permanently lower GDP growth 1/	31	11	7	6	6	5	8	9	5	5
B. Bound tests										
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	31	11	7	7	6	6	9	9	6	7
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	31	11	7	7	6	6	8	9	5	5
B3. Combination of B1-B2 using one half standard deviation shocks	31	11	7	7	6	6	8	9	5	4
B4. One-time 30 percent real depreciation in 2006	31	13	9	8	8	7	10	11	6	5
B5. 10 percent of GDP increase in other debt-creating flows in 2006	31	11	7	7	6	6	8	9	5	5

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Fig. 1.São Tomé and Príncipe: Indicators of Public Debt Under Alternative Scenarios, 2005-2025



Source: Staff projections.

Table 2a. São Tomé and Príncipe: External Debt Sustainability Framework, Baseline Scenario, 2005–2025 1/
(In percent of GDP, unless otherwise indicated)

	Est. 2005	Historical Average 5/	Standard Deviation 5/	Projections 2006	2007	2008	2009	2010	2005-10 Average	2011	2012	2015	2025	2011–25 Average
External debt (nominal) 1/	**419.7**			**355.4**	**328.1**	**301.3**	**275.2**	**249.2**		**224.2**	**54.9**	**16.3**	**19.9**	
Of which: public and publicly guaranteed (PPG)	419.7			355.4	328.1	301.3	275.2	249.2		224.2	54.9	16.3	19.9	
Change in external debt	-48.3			-64.3	-27.3	-26.8	-26.1	-26.0		-25.1	-169.3	-0.7	0.0	
Identified net debt-creating flows	-11.6			4.2	6.2	7.0	5.9	5.1		67.3	75.0	-39.6	10.7	
Non-interest current account deficit	**28.6**	**16.9**	**10.2**	**24.3**	**26.9**	**29.6**	**28.5**	**27.9**		**103.3**	**84.1**	**-36.6**	**16.3**	-2.6
Deficit in balance of goods and services	57.8			59.4	61.7	64.9	61.0	56.8		127.9	30.0	-71.4	-9.9	
Exports	30.5			34.9	35.2	35.2	34.3	34.1		34.7	47.0	93.4	50.1	
Imports	88.4			94.3	97.0	100.1	95.3	90.8		162.7	77.0	22.0	40.2	
Net current transfers (negative = inflow)	-29.0	-40.2	13.5	-34.4	-35.3	-34.6	-32.1	-29.0		-24.7	-5.9	-1.0	-1.0	-3.0
Other current account flows (negative = net inflow)	-0.3			-0.7	0.5	-0.6	-0.4	0.2		0.0	60.0	35.7	27.2	
Net FDI (negative = inflow)	**-5.0**	**-6.1**	**2.4**	**-5.4**	**-5.4**	**-5.5**	**-5.5**	**-5.5**		**-20.1**	**-5.2**	**-1.8**	**-4.2**	-4.2
Endogenous debt dynamics 2/	**-35.1**			**-14.7**	**-15.3**	**-17.2**	**-17.1**	**-17.3**		**-15.9**	**-3.9**	**-1.2**	**-1.4**	
Contribution from nominal interest rate	4.5			4.0	2.5	2.1	1.9	1.3		2.0	0.5	0.1	0.2	
Contribution from real GDP growth	-16.3			-18.7	-17.8	-19.4	-19.1	-18.6		-17.9	-4.4	-1.3	-1.6	
Contribution from price and exchange rate changes	-23.4			…	…	…	…	…		…	…	…	…	
Residual (3-4) 3/	**-36.7**			**-68.6**	**-33.4**	**-33.8**	**-32.0**	**-31.1**		**-92.4**	**-244.3**	**38.9**	**-10.7**	
NPV of external debt	152.6			110.1	106.4	101.6	96.7	92.2		85.6	21.6	6.9	10.2	
In percent of exports	499.5			315.5	301.9	288.7	282.1	270.8		246.4	45.9	7.4	20.3	
NPV of PPG external debt	**152.6**			**110.1**	**106.4**	**101.6**	**96.7**	**92.2**		**85.6**	**21.6**	**6.9**	**10.2**	
In percent of exports	499.5			315.5	301.9	288.7	282.1	270.8		246.4	45.9	7.4	20.3	
Debt service-to-exports ratio (in percent)	56.0			19.1	12.4	11.4	10.6	9.1		11.8	2.4	0.4	2.1	
PPG debt service-to-exports ratio (in percent)	56.0			19.1	12.4	11.4	10.6	9.1		11.8	2.4	0.4	2.1	
Total gross financing need (billions of U.S. dollars)	0.0			0.0	0.0	0.0	0.0	0.0		0.1	0.4	-0.6	0.1	
Non-interest current account deficit that stabilizes debt ratio	76.8			88.6	54.2	56.5	54.6	53.9		128.3	253.3	-35.9	16.3	
Key macroeconomic assumptions														
Real GDP growth (in percent)	3.8	3.0	1.1	4.5	5.5	6.5	7.0	7.5	6.2	8.0	8.0	8.0	8.0	8.0
GDP deflator in US dollar terms (change in percent)	5.3	1.2	7.1	-3.3	4.1	3.4	3.4	3.1	2.1	3.2	277.2	-4.0	-9.5	21.6
Effective interest rate (percent) 4/	1.0	1.2	0.5	1.0	0.8	0.7	0.7	0.5	0.7	0.9	0.9	0.9	0.9	0.9
Growth of exports of G&S (US dollar terms, in percent)	6.9	9.3	12.6	15.4	10.9	10.0	7.7	10.1	10.8	13.6	451.6	2.4	-14.7	54.3
Growth of imports of G&S (US dollar terms, in percent)	9.0	4.4	16.6	7.8	12.9	13.7	5.3	5.6	9.1	99.5	92.9	-27.3	10.1	17.1
Grant element of new public sector borrowing (including IMF; in percent)	...	...	...	42.4	44.0	47.1	48.9	46.4	45.8	48.1	48.1	48.1	48.1	48.1
Grant element of new public sector borrowing (excluding IMF; in percent)	...	...	...	46.9	47.6	49.3	48.9	46.4	47.8	48.1	48.1	48.1	48.1	48.1
Memorandum item:														
Nominal GDP (billions of US dollars)	0.1			0.1	0.1	0.1	0.1	0.1		0.1	0.5	1.6	1.1	

Source: Staff projections.

1/ External public debt only.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Current-year interest payments devided by previous period debt stock.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

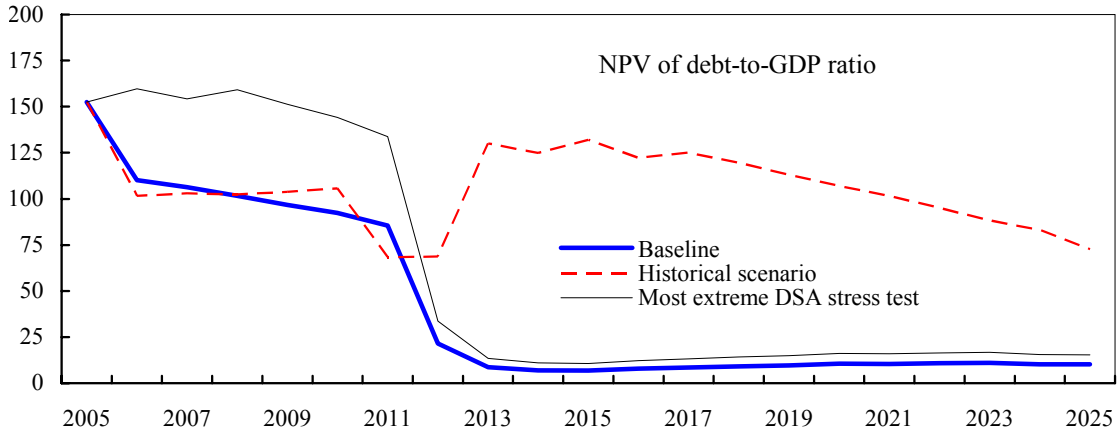
Table 2b. São Tomé and Príncipe: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005–2025
(In percent)

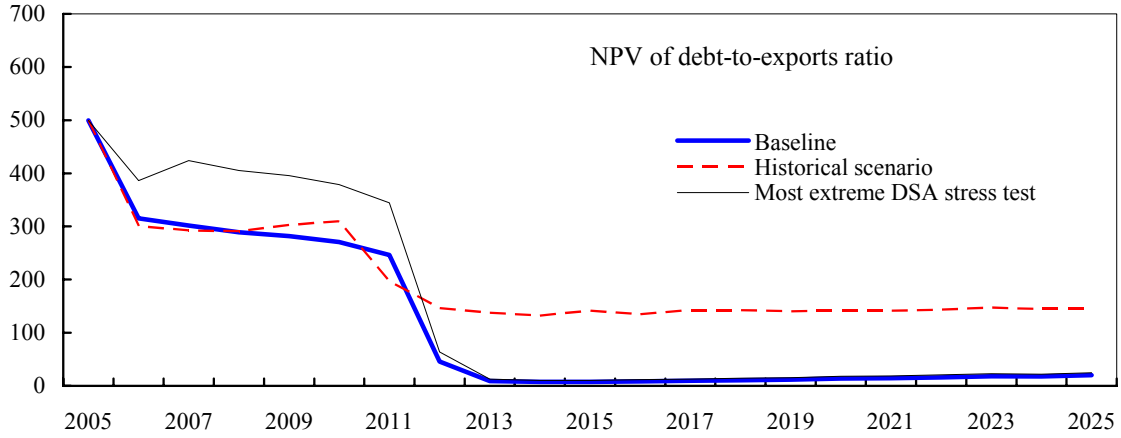
	Est.	Projections								
	2005	2006	2007	2008	2009	2010	2011	2012	2015	2025
					NPV of debt-to-GDP ratio 1/					
Baseline	153	110	106	102	97	92	86	22	7	10
A. Alternative Scenarios										
A1. Key variables at their historical averages in 2006-25 2/	153	102	103	102	104	106	68	69	132	73
A2. New public sector loans on less favorable terms in 2006-25 3/	153	110	108	105	101	98	92	23	8	13
A3. MDRI	…	61	62	61	60	59	55	14	5	7
B. Bound Tests										
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	153	114	114	109	103	99	91	23	7	11
B2. Export value growth at historical average minus one standard deviation in 2006-07 4/	153	113	109	104	99	94	87	22	7	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	153	114	122	116	111	106	98	25	8	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 5/	153	115	111	106	100	96	89	22	7	10
B5. Combination of B1-B4 using one-half standard deviation shocks	153	114	121	115	109	104	97	24	8	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 6/	153	160	154	159	151	144	134	34	11	15
					NPV of debt-to-exports ratio					
Baseline	500	315	302	289	282	271	246	46	7	20
A. Alternative Scenarios										
A1. Key variables at their historical averages in 2006-25 2/	500	301	293	291	303	310	197	146	141	145
A2. New public sector loans on less favorable terms in 2006-25 3/	500	315	307	298	296	287	265	50	8	27
A3. MDRI	…	176	176	174	175	172	158	30	5	14
B. Bound Tests										
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	500	326	304	291	284	273	248	46	7	20
B2. Export value growth at historical average minus one standard deviation in 2006-07 4/	500	386	424	405	395	379	344	64	10	25
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	500	327	304	291	284	273	248	46	7	20
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 5/	500	329	314	300	293	281	256	48	8	19
B5. Combination of B1-B4 using one-half standard deviation shocks	500	362	372	356	347	333	303	56	9	22
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 6/	500	457	302	312	304	292	266	49	8	21
					Debt service-to-export ratio					
Baseline	56.0	19.1	12.4	11.4	10.6	9.1	11.8	2.4	0.4	2.1
A. Alternative Scenarios										
A1. Key variables at their historical averages in 2006-25 2/	56.0	19.1	12.4	11.8	11.5	9.0	13.2	10.0	4.4	24.3
A2. New public sector loans on less favorable terms in 2006-25 3/	56.0	18.9	12.3	11.6	11.2	9.1	10.6	2.1	0.4	2.3
A3. MDRI	…	9.9	6.7	6.5	6.0	5.4	8.8	1.8	0.3	1.3
B. Bound Tests										
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	56.0	19.6	12.3	11.3	10.5	8.1	10.9	2.3	0.4	2.1
B2. Export value growth at historical average minus one standard deviation in 2006-07 4/	56.0	22.8	17.0	15.7	14.9	11.5	15.3	3.2	0.7	3.1
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	56.0	19.6	12.3	11.3	10.5	8.1	10.9	2.3	0.4	2.1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 5/	56.0	19.1	12.4	11.5	10.8	8.3	11.1	2.3	0.5	2.2
B5. Combination of B1-B4 using one-half standard deviation shocks	56.0	21.4	14.9	13.9	13.1	10.1	13.4	2.8	0.6	2.7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 6/	56.0	27.7	12.6	11.5	10.8	8.3	11.1	2.3	0.5	2.2
Memorandum item:										
Grant element assumed on residual financing (i.e., financing required above baseline) 7/	53	51	49	47	45	43	40	38	36	32

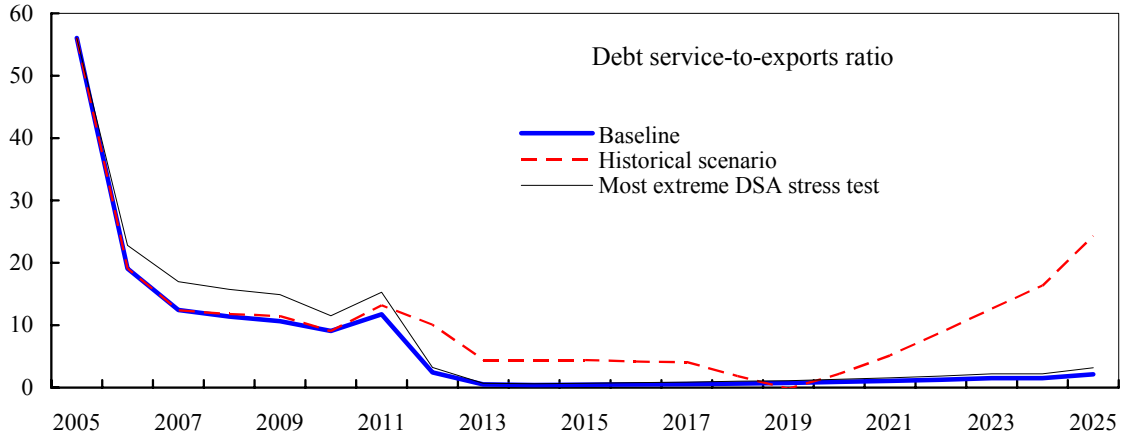
Source: Staff projections.

1/ By the year 2023 real GDP growth slows down significantly due to the projected decline in petroleum production.
2/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
4/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
5/ Includes official and private transfers and FDI.
6/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
7/ Based on historical level of concessionality. Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Fig. 2. São Tomé and Príncipe: Indicators of External Debt, 2005–25
(In percent)

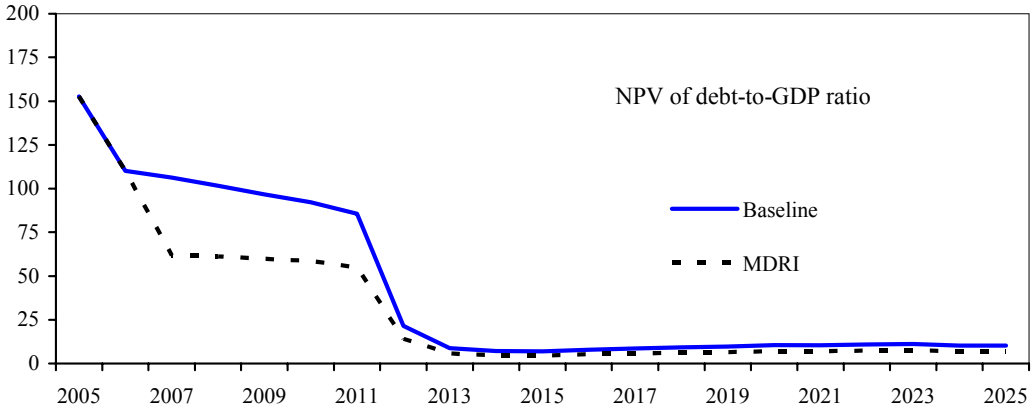


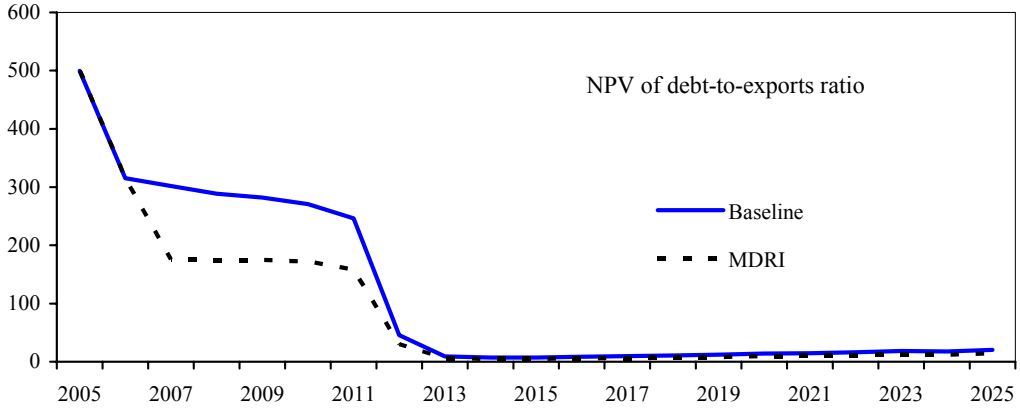


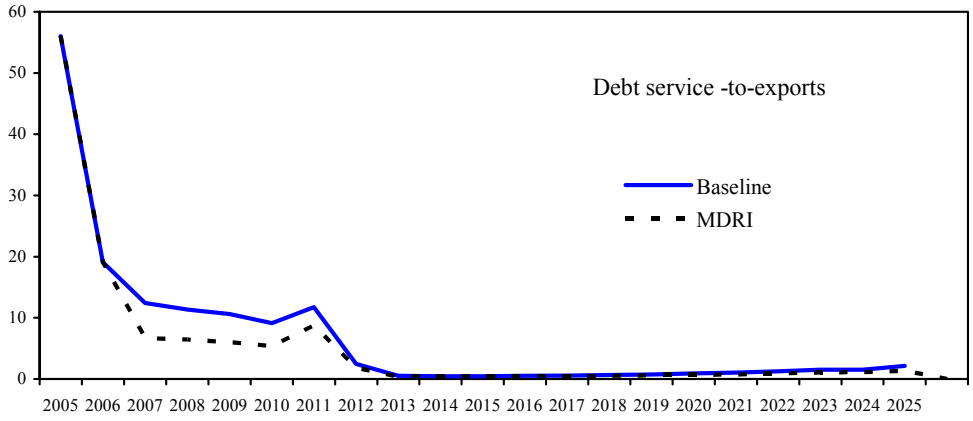


Source: Staff projections.

Fig. 3. São Tomé and Príncipe: External Debt under the Baseline and MDRI, 2005–2025
(In percent)







Source: Staff projections.

Table 3. São Tomé and Príncipe: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt under DMRI, 2005–2025
(In percent)

	Est.	Projections								
	2005	2006	2007	2008	2009	2010	2011	2012	2015	2025
					NPV of debt-to-GDP ratio 1/					
Baseline under DMRI	153	61	62	61	60	59	55	14	5	7
A. Alternative Scenarios										
A1. Key variables at their historical averages in 2006-25 2/	153	50	57	58	61	64	28	27	94	50
A2. New public sector loans on less favorable terms in 2006-25 3/	153	61	64	65	65	64	61	16	5	10
A3. MDRI	...	66	67	66	64	62	58	15	5	7
B. Bound Tests										
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	153	67	70	69	67	66	61	16	5	8
B2. Export value growth at historical average minus one standard deviation in 2006-07 4/	153	64	64	64	62	61	57	14	5	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	153	67	75	74	72	71	66	17	5	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 5/	153	66	66	65	64	62	58	15	5	7
B5. Combination of B1-B4 using one-half standard deviation shocks	153	66	72	71	70	68	64	16	5	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 6/	153	89	90	148	143	138	129	33	11	13
					NPV of debt-to-exports ratio					
Baseline under DMRI	500	176	176	174	175	172	158	30	5	14
A. Alternative Scenarios										
A1. Key variables at their historical averages in 2006–2025 2/	500	148	163	166	178	188	79	58	101	100
A2. New public sector loans on less favorable terms in 2006–25 3/	500	176	181	183	189	189	177	34	6	21
A3. MDRI	...	190	189	187	187	183	167	31	5	14
B. Bound Tests										
B1. Real GDP growth at historical average minus one standard deviation in 2006–07	500	192	186	184	184	181	167	31	5	14
B2. Export value growth at historical average minus one standard deviation in 2006–07 4/	500	220	251	248	248	244	224	42	7	17
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006–07	500	193	187	185	185	182	167	31	5	15
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006–07 5/	500	189	188	185	186	182	167	31	5	13
B5. Combination of B1-B4 using one-half standard deviation shocks	500	209	224	221	221	217	199	37	6	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 6/	500	255	176	291	287	278	255	48	8	18
					Debt service-to-export ratio					
Baseline under DMRI	56.0	9.9	6.7	6.5	6.0	5.4	8.8	1.8	0.3	1.3
A. Alternative Scenarios										
A1. Key variables at their historical averages in 2006–2025 2/	56.0	13.0	6.3	5.5	6.0	4.1	9.1	6.6	2.3	15.6
A2. New public sector loans on less favorable terms in 2006–2025 3/	56.0	12.8	6.5	5.9	6.7	5.3	7.6	1.4	0.3	1.5
A3. MDRI	...	9.9	6.7	6.5	6.0	5.4	8.8	1.8	0.3	1.3
B. Bound Tests										
B1. Real GDP growth at historical average minus one standard deviation in 2006–07	56.0	13.3	6.6	5.7	6.0	4.3	7.9	1.7	0.3	1.4
B2. Export value growth at historical average minus one standard deviation in 2006–07 4/	56.0	15.5	9.0	7.9	8.7	6.2	11.2	2.3	0.5	2.0
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006–07	56.0	13.4	6.7	5.7	6.1	4.3	7.9	1.7	0.3	1.4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006–07 5/	56.0	13.0	6.7	5.9	6.2	4.4	8.1	1.7	0.4	1.4
B5. Combination of B1-B4 using one-half standard deviation shocks	56.0	14.6	8.0	7.1	7.6	5.5	9.8	2.0	0.5	1.8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 6/	56.0	18.9	8.1	7.1	7.3	5.5	9.0	1.8	0.6	2.0
Memorandum item:										
Grant element assumed on residual financing (i.e., financing required above baseline) 7/	53	51	49	47	45	43	40	38	36	32

Source: Staff projections.

1/ By the year 2023 real GDP growth slows down significantly due to the projected decline in petroleum production.
2/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
4/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
5/ Includes official and private transfers and FDI.
6/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
7/ Based on historical level of concessionality. Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.